GOVERNMENT POLYTECHNIC, NAGPUR

(An autonomous institute of government of Maharashtra)

DEPARTMENT OF INFORMATION TECHNOLOGY

Optimizing Bird Classification via Neural Network Hyperparameter Tuning

Presented by:

Pratik Dhaktode - 2207017 [Group Leader]

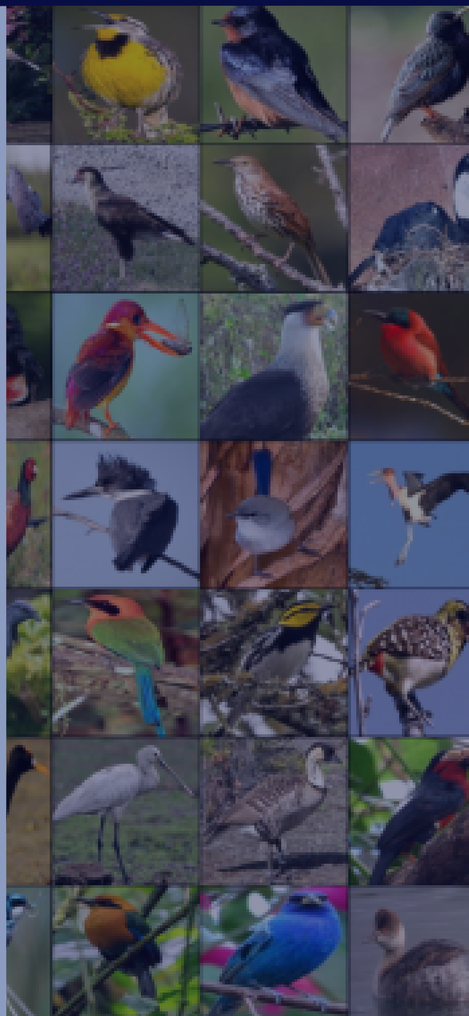
Raviraj Bhore - 2207006

Suhani Bighane - 2207007

Lavanya Vaidya - 2207063

Guided by:

Mr. R. L. Meshram



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INTRODUCTION



Bird classification

Birds classification refers to the process of identifying and categorizing different species of birds based on their physical features, sounds, or behaviors. In modern applications, this task is often done using computer vision and machine learning techniques.

Key Research Areas in Bird Classification

- **Taxonomy and Systematics:** Identifying species, understanding evolutionary relationships, and updating classification systems.
- **Biogeography:** Studying bird distribution, migration, and island diversity.
- **Behavior and Ecology:** Analyzing social interactions, foraging strategies, and habitat preferences.

Challenges while classifying birds

- **Subspecies & Hybridization**: Confusion from closely related forms.
- **Behavioral Differences**: Similar appearances, differing behaviors.
- **Geographic Variation**: Variations in appearance based on location complicate identification efforts.
- **Limited Access to Specimens**: Incomplete field guides hinder accurate identification.

Accurately classifying birds is challenging due to overlapping features, behavioral variations, and environmental influences.

Existing Work done on birds classification



Existing Methods:

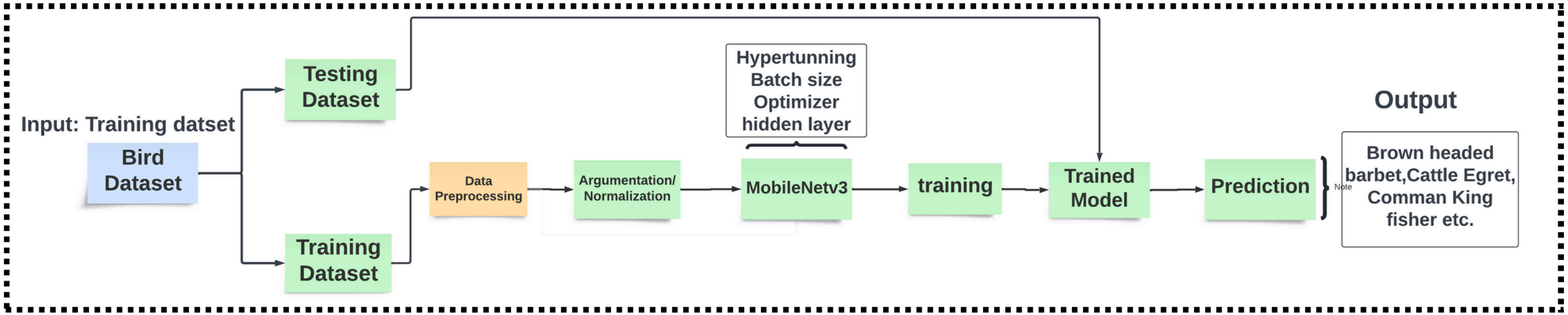
- **CNNs:** Ideal for bird classification due to their effectiveness in spatial feature extraction.
- **EfficientNet:** Balances accuracy and computational efficiency with compound scaling; EfficientNetB0 is effective for bird classification.
- **Transfer Learning:** Utilizes pretrained models (e.g., ResNet, MobileNetV3, VGG) to accelerate training and leverage features from larger datasets like ImageNet.
- **ResNet50 and MobileNet:** Popular for their strong feature extraction and lightweight architecture.

Result

- **EfficientNetB0:** [**AUTHOR 1**]:MINGXING TAN AND QUOC V. LE
- **Accuracy:** ~85-90%
- **Strengths:** High accuracy and efficiency due to balanced scaling.
- **ResNet50: [Author 2]**: Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun
- **Accuracy:** ~80-85%
- **Strengths:** Strong feature extraction with residual connections.
- **MobileNetV3**: [AUTHOR 3: ANDREW HOWARD, MARK SANDLER, GRACE CHU, LIANG-CHIEH CHEN]
- **Accuracy:** ~75-80%
- **Strengths:** Lightweight and efficient, but slightly lower accuracy compared to the other models.

METHODOLOGY

- **Concept Map**



In this project,we have used **indian bird classification(birds_25)** dataset . We have used techniques like rotating,flipping and resizing are applied to increase the dataset's variety. The performance i.s measured using metrics like accuracy and loss. We have evaluated, validated and tested the **mobilenetv3 model**. We have **hypertunned** the pretrained model mobilenetv3 by changing parameters like **optimizer, reducing hidden layer**,etc.The reports are generated in the form of confusion matrics and classification report. By observing the above things like changing parameters we get the results.

The technology used are:

MobileNetV3

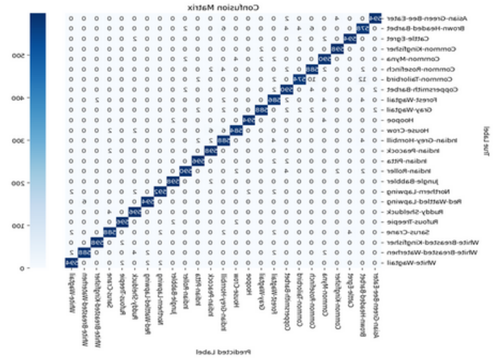
MACHINE LEARNING

1. MODEL OVERVIEW:

- MobileNetV3, introduced by Google, is an efficient and lightweight convolutional neural network (CNN) architecture, primarily designed for mobile and edge devices.
- It balances performance and computational efficiency, making it suitable for tasks like bird species classification.
- MobileNetV3 comes in two variants, MobileNetV3-Large and MobileNetV3-Small. For this dataset we have used MobileNetV3 large because it is optimized for high-performance tasks with slightly higher computational complexity and better accuracy.

2. PERFORMANCE METRIC:

- The model achieved **95.09% accuracy**, meaning it correctly classified 95.09% of the test samples.
- This high accuracy indicates the model's ability to recognize and classify bird species with a high level of reliability.



Confusion matrics

	precision	recall	f1-score	support
Asian-Green-Bee-Eater	0.99	0.99	0.99	600
Brown-Headed-Barbet	0.96	0.96	0.96	600
Cattle-Egret	0.99	0.99	0.99	600
Common-Kingfisher	0.99	1.00	0.99	600
Common-Myna	0.98	0.98	0.98	600
Common-Rosefinch	0.96	0.98	0.97	600
Common-Tailorbird	0.99	0.96	0.97	600
Coppersmith-Barbet	0.97	0.98	0.98	600
Forest-Wagtail	0.97	0.98	0.97	600
Gray-Wagtail	0.99	0.98	0.99	600
Hoopoe	1.00	0.99	0.99	600
House-Crow	0.99	0.97	0.98	600
Indian-Grey-Hornbill	0.96	0.98	0.97	600
Indian-Peacock	0.98	1.00	0.99	600
Indian-Pitta	0.99	0.99	0.99	600
Indian-Roller	0.99	0.98	0.99	600
Jungle-Babbler	0.99	1.00	1.00	600
Northern-Lapwing	0.99	0.99	0.99	600
Red-Wattled-Lapwing	0.99	0.99	0.99	600
Ruddy-Shelduck	0.98	0.99	0.99	600
Rufous-Treepie	0.99	0.99	0.99	600
Sarus-Crane	0.99	0.98	0.98	600
White-Breasted-Kingfisher	1.00	1.00	1.00	600
...				
accuracy			0.98	15000
macro avg	0.98	0.98	0.98	15000
weighted avg	0.98	0.98	0.98	15000

Classification report

Final Result using SGD Optimizer

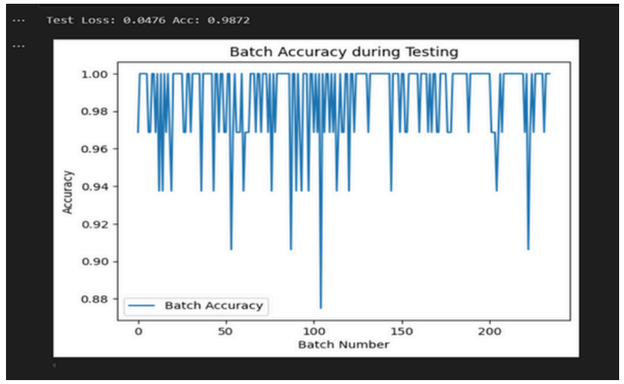
After experimenting with several optimizers, we selected Stochastic Gradient Descent (SGD).

Reason for using SGD:

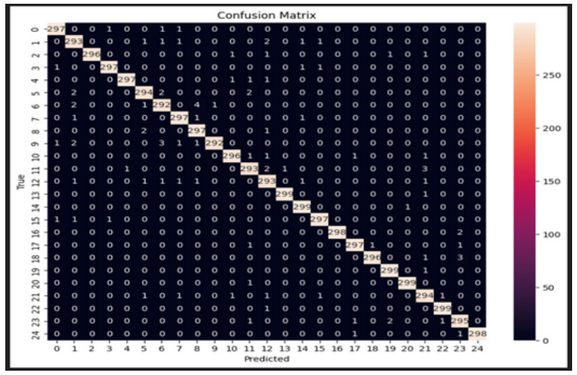
- In SGD, instead of using the entire dataset for each iteration, only a single random training example (or a small batch) is selected to calculate the gradient and update the model parameters.
- SGD helps in minimizing loss effectively over time.
- It allows fine control over learning with parameters like momentum and learning rate.

Effect on Performance:

- With the SGD optimizer, the model's performance improved significantly, reaching an **accuracy of 98.72%** with a loss of only 0.0476.
- This shows that SGD with momentum provided faster and more stable convergence, leading to better generalization on unseen data. SGD was thus the most effective optimizer among the ones tested.

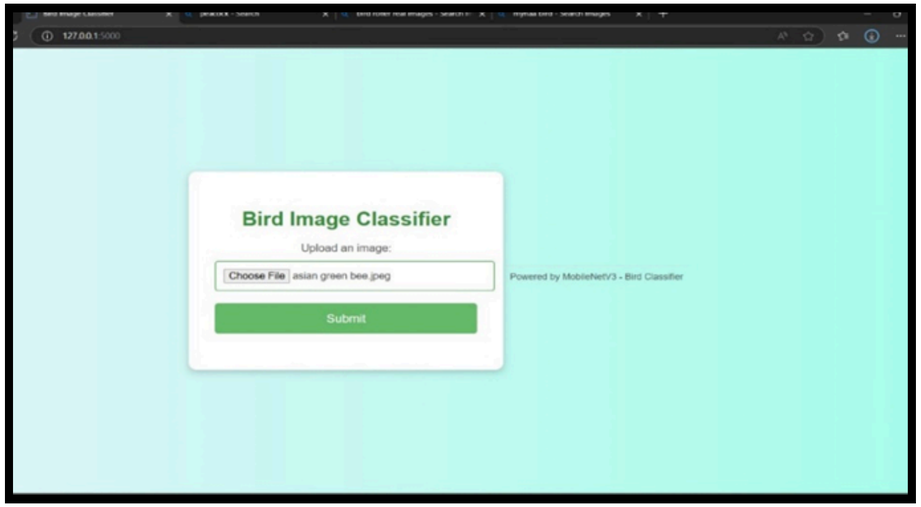


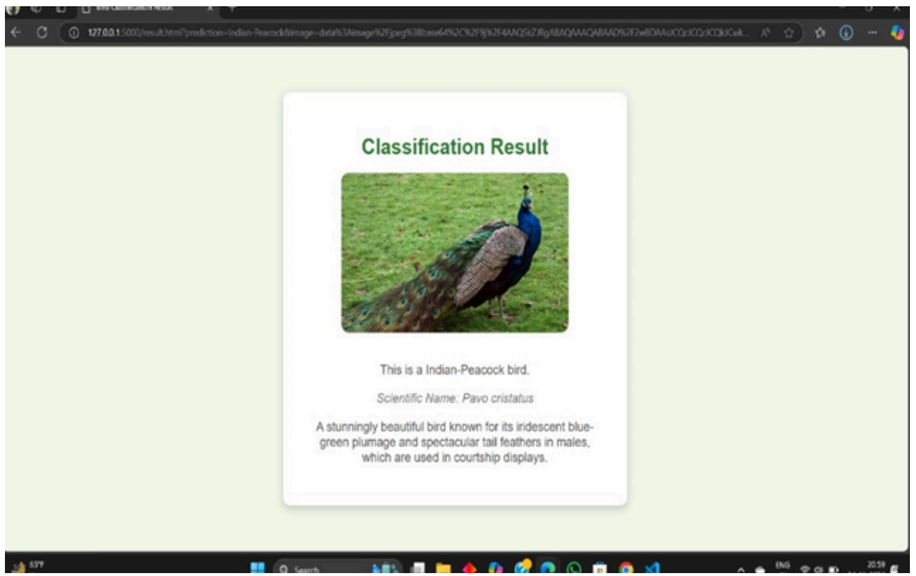
Accuracy graph



Confusion matrics

Final Result





We developed a website for bird species classification, where users can upload an image of a bird, and the system identifies the species and provides detailed information about it, including habitat, diet, and conservation status.The interface is powered by the MobileNetV3 model, ensuring quick and accurate predictions. This platform serves as a valuable tool for who are interestid in birds, researchers ,offering an accessible and efficient way to identify and learn about various bird species.

Conclusion



In this project, we successfully improved bird species classification using MobileNetV3. By fine-tuning the model, including reducing hidden layers, changing optimizers, and adjusting hyperparameters, we achieved better accuracy and efficiency.

- **Results:** The optimized model is faster and more effective at classifying birds.

- **Key Insight:** Simple adjustments and tuning can lead to significant performance improvements without making the model more complex.

Reference



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Thank You